

## 2025-2026 Academic Year Second Semester

### Machine Learning Course Project

#### Multimodal User Intention Recognition

##### 1. Project Background

With the successive releases and commercialization of next-generation augmented reality (AR) glasses products such as Apple Vision Pro, Meta Orion, XREAL Air, and Rokid AR Studio, AR devices are rapidly moving from laboratory prototypes to the consumer and enterprise markets. AR glasses are widely regarded as the next-generation general-purpose computing and interaction platform after smartphones. However, unlike the traditional PC mouse and keyboard and mobile phone touchscreen interaction paradigms, AR glasses have typical characteristics such as "no controller, no screen touch control, spatialization, mobility, and long-term wear," which makes traditional single-modal interaction methods, such as pure voice commands or pure gesture recognition, insufficient to meet the complex, natural, and efficient human-computer interaction needs. Single-modal interaction has significant shortcomings in different scenarios, and multimodal fusion interaction has become the industry consensus. How to enable AR glasses to understand the user's true intentions has become one of the core bottlenecks restricting its large-scale popularization.

**2. Project Requirements:** Based on a given multimodal user interaction intent recognition model, improve the intent recognition model to address modal noise and missing data issues.

##### Implementation Content:

- 1) Divide the dataset into training and testing datasets. Use the interaction data of user A and user B as training data, and the interaction data of user C as testing data.
- 2) The given model code first extracts and saves features from the dataset, then calls the feature file for multimodal fusion. This project requires students to refactor the existing model code, reconstructing the training and testing code into an end-to-end model with raw data as input and intent category as output. Integrate the data preprocessing and feature processing code into the training and testing code, and provide screenshots in the report demonstrating that the end-to-end model code runs successfully.
- 3) After refactoring the code, construct a modal noise baseline. Add noise to each single-modal raw dataset at 20%, 40%, and 60% respectively, train the model, and test it on the test set to obtain a set of classification accuracy results.
- 4) Based on the refactored code, construct a modal missing data baseline. Discard any single modality and two modality datasets respectively, train the model, and test it on the test set to obtain a set of classification accuracy results.

5) In the code refactoring, improve the model by introducing new modules or loss terms to improve classification accuracy under modal noise and modality missing conditions.

### **3. Project report**

(1) No fewer than 15 citations. English citations should use the IEEE citation format, and Chinese citations should use the Shenzhen University Journal of Technology citation format. For journal articles, the content should include: "Serial Number, Author, Title, Journal Name, Year, Volume, Issue Number"; for books, the content should include: "Serial Number, Author, Title, Publisher, Publication Date, Page Number"; for online resources, the content should include: "Serial Number, Author, Title, URL" (if the online materials have been officially published, it is best to indicate them as journals or books).

(2) Specify the methodological improvements and innovations in the report.

(3) Submit a project report, specifying the tasks and contribution percentages of each team member.

(4) Group project defense (Week 17 experimental course): Each group will have 5 minutes for presentation + 2 minutes for Q&A.

### **4. Scoring Criteria**

(1) A reasonable and innovative method, with bonus points for novel innovative aspects.

(2) The project report should be logically clear, detailed, and accurately describe the methods, with sufficient experiments and rich results presentation. AI-generated project reports are not allowed.

(3) Individual scores are determined by the project score and the proportion of individual contribution.

## 2025-2026 学年第二学期

### 机器学习课程项目

#### 多模态用户交互意图理解

##### 1、项目背景

随着以 Apple Vision Pro、Meta Orion、XREAL Air、Rokid AR Studio 等为代表的新一代增强现实 (AR) 眼镜产品的相继发布与商业化落地, AR 设备正从实验室原型加速迈向消费级与企业级市场。AR 眼镜被广泛认为是继智能手机之后的下一代通用计算与交互平台。然而, 与传统的 PC 鼠标键盘、手机触屏交互范式不同, AR 眼镜的使用场景具有“无手柄、无屏幕触控、空间化、移动化、长时佩戴”等典型特征, 这使得传统的单一模态交互方式, 如纯语音命令或纯手势识别, 已难以满足复杂、自然、高效的人机交互需求。单一模态在不同场景下均存在显著短板, 多模态融合交互已成为业界共识的发展方向。如何让 AR 眼镜读懂用户的真实意图, 已成为制约其大规模普及的核心瓶颈之一。

##### 2、项目需求

在给定的多模态用户交互意图识别模型上, 针对模态噪声和缺失问题, 改进意图识别模型。

###### 实现内容:

1) 划分数据集为训练数据集和测试数据集, 使用用户 A 和用户 B 的交互数据作为训练数据, 使用用户 C 的交互数据作为测试数据。

2) 给定的模型代码是先对数据集进行特征提取、保存特征, 再调用特征文件进行多模态融合。本项目需要同学们重构已有模型代码, 将训练代码和测试代码重构成 raw data 为输入、意图类别为输出的端到端模型。将数据预处理和特征处理代码整合到训练代码和测试代码中, 需在报告中截图证明端到端模型代码跑通。

3) 重构代码后, 构建模态噪声基线。分别对各个单模态生数据进行 20%、40%、60% 加噪处理, 训练模型, 在测试集上测试, 得到一组分类精度结果。

4) 在重构代码上, 构建模态缺失基线。分别丢弃任意单个模态、两个模态生数据, 训练模型, 在测试集上测试, 得到一组分类精度结果。

5) 在重构代码上, 改进模型, 引入新模块或损失项, 改进模态噪声和模态缺失下的分类精度。

### 3、项目报告

- (1) 文献引用数量不少于 15 篇。英文文献引用格式使用 IEEE 文献引用格式，中文文献引用格式使用深大理工学报文献引用格式。文献是期刊时，内容有：“序号、作者、文献题目、期刊名、年份、卷号、期号”；文献是著作时，内容有：“序号、作者、书名、出版单位、出版年月、页码”；文献是网络资源时，内容有：“序号、作者、文献题目、网址”（若网上搜集的资料已正式出版或发表，最好以期刊和著作标注）。
- (2) 在报告中写明方法改进和创新点。
- (3) 提交项目报告，在项目报告中写明每个队员负责的任务和贡献比例。
- (4) 小组项目答辩（Week 17 实验课）：每个小组 5 分钟演讲+2 分钟问答。

### 4、评分标准

- (1) 方法合理且创新，方法有较新颖的创新点加分。
- (2) 项目报告写作逻辑清晰、叙述详细，方法描述详细准确，实验充分，结果展示丰富。  
AI 生成项目报告是不被允许的。
- (3) 个人得分由项目得分和个人贡献比例决定